

Banking System Project Documentation

Project Name: BSNB - Bhumi Shah National Bank

Description:

BSNB is a full-stack banking system web application developed using the MERN stack. It includes features such as user registration, login with JWT authentication, bank account creation and deletion, money transfers between accounts, transaction logging via triggers, and viewing account or transaction details.

Tech Stack:

- Frontend: React.js, TailwindCSS
- Backend: Node.js, Express.js
- Database: MySQL with Triggers
- Authentication: JWT
- Password Security: Bcrypt
- Image Assets: Managed in React /src/assets

Core Features:

1. User Management:

- Register, login (JWT), update, and delete users.
- Routes: /users

2. Account Management:

- Create, read, and delete bank accounts.
- Routes: /accounts

3. Transfers:

- POST /transfers/transfer-money
- Transfers money between two valid account IDs.
- Includes validations, atomic transactions (START TRANSACTION, COMMIT, ROLLBACK).

4. Transactions:

- Automatic logging of deposits and withdrawals via MySQL triggers.
- Manual logging for transfers.
- Route: /transactions/add and /transactions

5. Parameterized Queries:

- Used throughout to prevent SQL injection.

6. Error Handling:

- Errors during transactions roll back changes and return meaningful messages.
- Validations include account existence, sufficient balance, and data completeness.

7. UI/UX:

- Responsive design using TailwindCSS.
- Landing page with a carousel, animated feature cards.
- Separate login and register pages.

API Routes Summary:

Method	Endpoint	Description
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POST	/users/create	Register a new user	
GET	/users/read/:email	Login & get JWT token	
PUT	/users/update	Update user details	
DELETE	/users/delete/:user_id	Delete user	
GET	/users/count	Get user count	
POST	/accounts/create	Create a bank account	
GET	/accounts/read	Read all accounts	
DELETE	/accounts/delete	Delete an account	
POST	/transfers/transfer-money	Transfer money between accounts	
POST	/transactions/add	Add manual transaction log	
GET	/transactions	Get all transactions	

Routing Setup:

- app.use('/users', userRoutes);
- app.use('/accounts', accountRoutes);
- app.use('/transactions', transactionRoutes);
- app.use('/transfers', transferRoutes);

Conclusion:

The BSNB banking system demonstrates a scalable and secure backend integrated with a modern frontend for a complete banking experience. It simulates real-world banking operations while ensuring data integrity and security.